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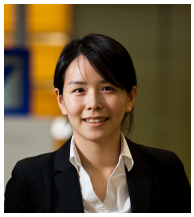

Ying Siu (Sheryl) Liang

Research Interests

I am interested in developing generalisable solutions for robots to operate with end-users and in real-world applications. I have developed solutions for end-user robot programming by demonstration and object permanence to control robot behaviour in human. My research interests include, but are not limited to: Artificial Intelligence, Cognitive Robotics, End-user Robotics, Learning from Demonstration, Human-Robot Interaction.

Education

2016–2019 **Ph.D. in Computer Science**, *Université Grenoble Alpes*, France.
Major in Artificial Intelligence and Robotics. Defended in June 2019
Dissertation: *End-User Robot Programming of Action Models for Symbolic Task Planning*

2015–2016 **Master of Science (M.S) in Informatics**, *Grenoble INP - ENSIMAG*, France.
Specialisation: Artificial Intelligence and Web
Courses included: Machine Learning, Natural Language Processing, Multi-Agent Systems

2008–2012 **Integrated Master of Science (MSci) in Mathematics and Computer Science**, *Imperial College London*, United Kingdom, *First Class Honours*.
Courses included: Software Engineering Algorithms, Programming & Data Structures

2000–2008 **Bundesrealgymnasium 18, Wien**, *Vienna*, Austria, Secondary school with emphasis on Geometric Modelling, Information and Communication Technology, *GPA 1.0*.
Matura (A-Level equivalent) in Mathematics, English, Geometric Modelling, German, French

Skills

Development	C++, Python, Clojure, JavaScript, HTML, LaTeX, Git
Previous	Java, Prolog, Haskell, MATLAB, C#, Perl, PHP, SQL
OS	Ubuntu Linux, Windows, MacOS
	ROS Indigo, MoveIt, Visual Studio Code, PCL, OpenCV, Polymer
Robots	Universal Robot (UR5), Baxter Research Robot, Fetch Mobile Manipulator, PR2
Spoken languages	German (native), English (bilingual), French (C2), Chinese (C1), Spanish (A2)

Other Courses

Q3 2021	IS5452 Affective Computing, National University of Singapore, Singapore	
Oct 2018	Deep RL practical session using PyTorch, Criteo AI Lab, France	
Aug 2018	ROS Summer School, FH Aachen, Germany	*OpenCV, SLAM*

Research and Work Experience

Jan 2020–Present **Research Scientist**, *Institute for High Performance Computing*, A*STAR, Singapore.
CollabAI project in Social and Cognitive Computing (Clojure, Python, ROS melodic)

- Worked on the robot's cognitive architecture (ICARUS) which combines input from perception and NLP to perform high-level reasoning and task planning to achieve human-robot collaborative goals with a Universal Robot (UR5, UR3e).
- Developed an object permanence model (AAPA) that tracks carried objects and invisible displacements and outperformed state-of-the-art models.

Jun–Sep 2019 **Research Intern**, *Institute for Infocomm Research*, A*STAR, Singapore.
Inferring the Geometric Nullspace for Robot Skills from Human Demonstrations
- Worked on a setup to infer geometric constraints and their nullspace from human demonstrations (using OptiTrack) that can be used by the iCub robot to generalise skills to perform more complex tasks (C++).

Oct 2016 – Jun 2019 **Ph.D. candidate**, *Grenoble Informatics Lab*, Univ. Grenoble Alpes, France.
End-User Robot Programming of Action Models for Symbolic Task Planning
- Proposed and implemented an interactive robot programming framework as an end-to-end software in ROS which allowed end-users to teach a Baxter robot new actions from scratch.
- My solution combined Programming by Demonstration and Automated Planning techniques to simultaneously learn low- and high-level actions (C++, ROS indigo).
- The developed system combines solutions from object perception (PCL), motion planning (MoveIt), task planning (PDDL) and human–robot interaction (Web App in JavaScript/Polymer following HCI concepts).
- I conducted human-robot experiments for user acceptance testing and system evaluation.

Jul–Dec 2017 **Research Intern**, *Human-Centered Robotics Lab*, Univ. of Washington, USA.
Simultaneous End-User Programming of Goals and Actions for Robotic Shelf Organisation
- Proposed and implemented a system on a Fetch mobile manipulator to learn goals and actions from human demonstrations for shelf organising tasks (C++, ROS indigo, PCL).
- Derived a goal inference method to accelerate the programming process, with priors based on a dataset of 4000 grocery shelf images.
- Evaluated the method with simulated teaching strategies and a user study on Amazon Mechanical Turk (Python, JavaScript).

Feb–Jun 2016 **Research Intern**, *Grenoble Informatics Lab*, Univ. Grenoble Alpes, France.
Robot Programming by Demonstration in Cobotic Environments
- Built and evaluated a prototype of a robot programming framework with a Baxter robot for object detection and action execution (Python, OpenCV, ROS indigo).
- Conducted qualitative user experiments to evaluate the framework's usability.

Sep 2012 – Sep 2015 **Software Engineer**, *Global Technology, Deutsche Bank*, London, United Kingdom.
- Developed functionalities for data retrieval from external data sources to allow users to query and modify data directly from an OLAP Cube via a web app (C#, HTML/JavaScript).

Additional Responsibilities

April 2021 **Co-coordinator**, *NIE-ASTAR AI in Education workshop*, Singapore.

2017– **Reviewer**, *IROS'21, ICRA'18,19,21, ACII'21, RSS'19, RO-MAN'19, HRI'18*.

May–Jul 2018 **Supervision of 2 undergraduate interns**, *UI design*, Grenoble, France.

Feb 2017 **Teaching Assistant**, *Introduction to Databases*, Grenoble, France.

2021– **Math Tutor Volunteer**, *South Central Community Family Service Centre*, Singapore.

2020– **Trishaw Pilot Volunteer**, *Cycling Without Age*, Singapore.

2013–2015 **Math Tutor Volunteer**, *Volunteering Matters*, London, United Kingdom.

Awards and Honours

2016–2019 French Ministry PhD research grant by MSTII Doctoral School (ED MSTII)

Mar 2018 Prize for best presentation at the LIG PhD day

Jul 2017 IDEX Intl. Mobility Grant to collaborate with the Univ. of Washington

Jun 2017 Prize for best presentation at the ED MSTII PhD day

References

Dongkyu Choi (Dr), Senior Scientist, IHPC, choi_dongkyu@ihpc.a-star.edu.sg

Maya Cakmak (Dr), A. Professor, Univ. of Washington, mcakmak@cs.washington.edu

Publications and Other Communication

D. Choi, W. Shi, Y.S. Liang, K.H. Yeo, J.J. Kim (2021). **Controlling Industrial Robots with High-level Verbal Commands.** Intl. Conf. on Social Robotics (ICRS). Singapore: IEEE/RSJ Press. (*Accepted*)

Y.S. Liang, C. Zhang, D. Choi, K. Kwok (2021). **Improving Object Permanence using Action Annotations and Reasoning.** Intl. Conf. on Intelligent Robots and Systems (IROS). Virtual (Oral): IEEE/RSJ Press.

Y.S. Liang, D. Choi, K. Kwok (2021). **Maintaining a Reliable World Model using Action-aware Perceptual Anchoring.** Intl. Conf. on Robotics and Automation (ICRA). Xi'An, China (Oral): IEEE Press.

Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2021). **iRoPro, an interactive Robot Programming Framework.** Intl. Journal of Social Robotics (IJSR). Springer-Verlag.

C. Cai, Y.S. Liang, N. Somani, Y. Wu (2020). **Inferring the Geometric Nullspace of Robot Skills from Human Demonstrations.** Intl. Conf. on Robotics and Automation (ICRA). Virtual: IEEE Press.

Y.S. Liang (2019). **End-User Robot Programming in Cobotic Enviroments.** PhD thesis. Grenoble, France.

Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2019). **End-User Programming of Low- and High-Level Actions for Robotic Task Planning.** Intl. Symposium on Robot and Human Interactive Communication (RO-MAN). New Delhi, India (Oral): IEEE Press.

Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty, M. Cakmak (2018). **Simultaneous End-User Programming of Goals and Actions for Robotic Shelf Organization.** Intl. Conf. on Intelligent Robots and Systems (IROS). Madrid, Spain (Oral): IEEE/RSJ Press.

Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2017). **Evaluation of a Robot Programming Framework for Non-Experts using Symbolic Planning Representations.** Intl. Symposium on Robot and Human Interactive Communication (RO-MAN). Lisbon, Portugal (Poster): IEEE Press.

Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2017). **A Framework for Robot Programming in Cobotic Environments: First User Experiments.** Intl. Conf. on Mechatronics and Robotics Engineering (ICMRE). Paris, France (Oral): ACM.

May 2018 **Spotlight talk**, *A Robot Programming Framework for Non-Experts*, Robots in Human Environments (RHUM) Workshop, Grenoble, France.

Jun 2017 **Oral presentation**, *Robot Programming by Demonstration*, Affects, Artificial Companions and Interactions (ACAI) Working Group, Paris, France.